AI-Driven Retail Demand Forecasting and Inventory Recommendation Using Hybrid Time-Series Model Selection

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***Abstract*—Accurate demand forecasting is an important component of effective retail inventory management, as both overstocking and understocking can lead to significant operational and financial losses. Small and medium-sized retailers often rely on manual estimation, historical averages, or spreadsheet-based methods, while advanced enterprise forecasting systems can be expensive and complex. This paper presents SalesSense, an AI-powered retail demand forecasting and inventory recommendation platform designed to provide accessible and explainable forecasting capabilities. The proposed system evaluates three complementary forecasting approaches: Seasonal Autoregressive Integrated Moving Average (SARIMA), Prophet, and Long Short-Term Memory (LSTM). Rather than relying on a single forecasting technique for all products, SalesSense independently evaluates the three models for each store-product series using chronological backtesting and performance metrics including weighted Mean Absolute Percentage Error (wMAPE), Root Mean Square Error (RMSE), and the coefficient of determination (R^2). The best-performing model is selected as the primary forecasting model, while comparative results remain available to the user for transparency. The system uses the FreshRetailNet-50K dataset, consisting of hourly retail sales data across multiple stores and products, and incorporates contextual features such as discounts, holidays, activities, and weather conditions. A three-tier architecture comprising a React frontend, FastAPI backend, Python forecasting engine, and MongoDB database is proposed. The resulting platform aims to bridge the gap between basic spreadsheet-based forecasting and complex enterprise systems by providing actionable demand forecasts, confidence intervals, model transparency, and reorder recommendations through a user-friendly interface.**

***Index Terms*—demand forecasting, retail analytics, inventory management, SARIMA, Prophet, LSTM, time-series forecasting, machine learning, retail demand prediction**

I. Introduction

Retail businesses must continuously estimate future product demand to determine appropriate inventory levels and purchasing decisions. A retailer that orders significantly more inventory than required may face increased storage costs, capital being tied up in inventory, product spoilage, and markdowns. Conversely, insufficient inventory can result in stockouts, lost sales, dissatisfied customers, and emergency replenishment costs.

Traditional approaches used by small and medium-sized retailers frequently depend on intuition, basic averages, or spreadsheet calculations. Although these approaches are relatively simple, they may perform poorly when demand contains seasonal patterns, promotional effects, holidays, irregular fluctuations, or other contextual dependencies. At the opposite end of the spectrum, enterprise forecasting platforms provide sophisticated analytical capabilities but may introduce significant cost and complexity for smaller organizations.

SalesSense is designed to occupy the space between these two approaches by providing an accessible forecasting platform that evaluates multiple forecasting techniques and selects the model that performs best for a particular product. Instead of assuming that one forecasting technique is suitable for every product, the system evaluates different models against the same historical data.

A key design principle of SalesSense is that no single forecasting model is expected to perform optimally for every retail product. Stable products with strong seasonal patterns may be effectively modeled using classical statistical techniques, while products influenced by holidays, promotions, or nonlinear demand patterns may benefit from other approaches. The proposed system therefore incorporates SARIMA, Prophet, and LSTM and evaluates them on the same historical series.

The main objectives of the proposed system are:

1) Processing historical retail sales data.
2) Training multiple forecasting models for individual store-product combinations.
3) Comparing models using standardized evaluation metrics.
4) Selecting the best-performing forecasting model.
5) Generating future demand predictions with confidence information.

6) Translating demand predictions into inventory reorder recommendations.
7) Presenting forecasting results through an accessible web dashboard.

The proposed system uses the FreshRetailNet-50K dataset, which provides hourly retail sales data across 898 stores and 865 stock-keeping units (SKUs). For the current project scope, a subset of approximately 5–10 store-product series is planned for model development and evaluation.

II. Related Work

Time-series forecasting has traditionally relied on statistical models capable of exploiting temporal dependencies and recurring patterns in historical observations. SARIMA is a classical forecasting technique that incorporates autoregressive, differencing, moving-average, and seasonal components. In SalesSense, SARIMA serves as an important baseline because it is relatively computationally inexpensive and can provide statistically grounded forecasts for stable and seasonal demand patterns.

Prophet provides an alternative forecasting approach based on decomposing a time series into components such as trend and seasonality while allowing the inclusion of calendar-related effects. This characteristic is particularly relevant to retail applications, where demand may change according to weekends, holidays, promotional activities, and other calendar events. The SalesSense design therefore includes holiday, weather, and promotional regressors within the Prophet configuration.

Deep learning approaches provide another perspective on demand forecasting. Long Short-Term Memory networks are recurrent neural network architectures designed to learn temporal dependencies in sequential data. Within SalesSense, LSTM is intended to capture nonlinear demand relationships and longer-term dependencies that may not be adequately represented by classical statistical models. However, the model requires more historical data and computational resources than simpler approaches.

The methodology referenced by the project includes the work of Ecevit et al. on short-term sales forecasting using LSTM- and Prophet-based models [1]. SalesSense adopts the general idea of evaluating multiple forecasting approaches on retail demand data while extending the design with SARIMA and a model-selection layer.

The key distinction of SalesSense is therefore not the use of an individual forecasting algorithm, but the integration of multiple forecasting approaches into a common evaluation and decision-making framework. Instead of presenting a single model as universally suitable, the system evaluates the models for each store-product series and provides evidence for the selected model.

III. Problem Statement and Objectives

A. Problem Statement

Retail demand varies significantly between products and over time. Conventional forecasting methods based on simple averages or manual judgment may fail to account for seasonality, promotions, holidays, weather effects, and nonlinear patterns. At the same time, enterprise-grade forecasting solutions may not be practical for small and medium-sized retailers due to their cost and complexity.

There is therefore a need for an accessible forecasting system that can:

- process historical retail sales data;
- capture different types of demand patterns;
- identify an appropriate model for each product;
- provide interpretable forecasting evidence; and
- translate predicted demand into practical inventory recommendations.

B. Objectives

The main objectives of SalesSense are:

1) **Demand Prediction:** Generate short-term demand forecasts for individual store-product combinations.
2) **Multi-Model Forecasting:** Implement SARIMA, Prophet, and LSTM models within a common forecasting framework.
3) **Model Evaluation:** Compare forecasting models using R^2, RMSE, and wMAPE.
4) **Automatic Model Selection:** Select the best-performing model for each store-product series.
5) **Explainability:** Show users why a particular model was selected based on its evaluation performance.
6) **Inventory Support:** Convert demand forecasts into recommended reorder quantities using a configurable safety margin.
7) **Visualization:** Provide historical trends, forecast charts, KPIs, and model comparisons through a web dashboard.
8) **Reporting:** Allow forecast results to be exported for further business analysis.

IV. Proposed System

A. System Overview

SalesSense is proposed as a standalone three-tier web application consisting of a React frontend, FastAPI backend, Python-based forecasting engine, and MongoDB data layer. The frontend provides the user interface, while the backend manages requests, authentication, data retrieval, and orchestration of forecasting operations. The forecasting engine performs preprocessing, model training, prediction, backtesting, and evaluation. MongoDB stores sales data, forecasts, metrics, and user configuration.

The overall processing flow is:

Historical Sales Data $\rightarrow$ *Preprocessing* $\rightarrow$ *Model Training* $\rightarrow$ *Forecasting* $\rightarrow$ *Backtesting* $\rightarrow$ *Model Evaluation* $\rightarrow$ *Best Model Selection* $\rightarrow$ *Demand Forecast* $\rightarrow$ *Inventory Recommendation*

A user configures a forecast request through the React interface. The FastAPI backend retrieves historical sales data from MongoDB and triggers the forecasting process. The forecasting engine preprocesses the data, executes the three

forecasting models, performs backtesting, evaluates the models, and returns the results. The forecast and evaluation metrics are then stored and presented through the dashboard.

B. Data Source

SalesSense uses the FreshRetailNet-50K dataset as its primary data source. The dataset contains hourly retail sales information covering 898 stores and 865 SKUs. The project uses a curated subset of approximately 5–10 store-product series to maintain a manageable computational scope within the academic project timeline [2].

The primary features identified for the forecasting pipeline include:

- `sale_amount`
- `hours_sale`
- `discount`
- `holiday_flag`
- `activity_flag`
- `precpt`
- `avg_temperature`
- `avg_humidity`
- `avg_wind_level`

These features allow the forecasting system to consider not only historical sales but also contextual factors that may influence demand.

C. Data Preprocessing

Retail sales data can contain missing observations and zero-sales periods. SalesSense does not automatically discard zero-sale hours because they may represent genuine periods of zero demand or potential stockout situations. Instead, these observations are retained during preprocessing.

Categorical variables such as holiday and activity indicators are encoded before being provided to forecasting models. Additional temporal characteristics such as day-of-week information are incorporated into the preprocessing process.

The dataset is divided chronologically into training and testing portions rather than using a random split. The proposed methodology specifies an 80% training and 20% testing split. A chronological split is particularly important for time-series forecasting because randomly mixing past and future observations can introduce information from the future into model training.

V. Forecasting Methodology

A. SARIMA

Seasonal Autoregressive Integrated Moving Average (SARIMA) is used as the classical statistical forecasting model.

SARIMA is particularly suitable for stable demand series that exhibit identifiable seasonal behavior. Within SalesSense, the $(p, d, q)(P, D, Q, s)$ parameters are grid-searched using Akaike Information Criterion (AIC). The seasonal period is selected according to the relevant daily or weekly pattern in the hourly sales data.

The primary advantages of SARIMA within the proposed system are:

- suitability for structured time-series data;
- relatively low computational cost;
- effectiveness on stable seasonal patterns;
- interpretability as a statistical baseline; and
- availability of statistically grounded confidence intervals.

B. Prophet

Prophet is used to model demand through trend, seasonality, and external effects. The proposed SalesSense configuration includes holiday information, weather-related variables, promotional information, linear growth, and logistic growth.

This makes Prophet particularly relevant to retail demand where calendar effects and external events can influence product sales. For example, a product may experience increased demand during holidays or promotional campaigns. Prophet's decomposition-based structure provides an interpretable representation of these effects.

C. LSTM

Long Short-Term Memory (LSTM) is included to capture nonlinear temporal relationships and longer-term dependencies in retail demand.

The planned architecture uses a bidirectional LSTM followed by global pooling and dense layers, with a 24-hour lookback window and early stopping. The architecture is based on the methodology referenced by Ecevit et al. [1].

LSTM is expected to be particularly useful for demand series containing complex patterns that are difficult to model using classical statistical techniques. However, it generally requires more data and computational resources than SARIMA or Prophet.

VI. Model Selection and Evaluation

A central feature of SalesSense is its model-selection mechanism. For every store-product series, SARIMA, Prophet, and LSTM are independently trained using the same historical data and evaluated against the same chronological test set. This ensures that model comparison is performed under consistent conditions.

The system evaluates the models using three metrics.

A. Root Mean Square Error

RMSE measures the magnitude of prediction errors while assigning greater importance to larger errors:

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}. \tag{1}$$

where y_i represents the actual demand and $\hat{y}_i$ represents the predicted demand.

B. Coefficient of Determination

R-squared indicates how effectively the model explains the variation in the observed data:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}. \tag{2}$$

where $\bar{y}$ represents the mean observed demand.

C. Weighted Mean Absolute Percentage Error

wMAPE is used as the primary model-selection criterion:

$$\text{wMAPE} = \frac{\sum_{i=1}^{n}|y_i - \hat{y}_i|}{\sum_{i=1}^{n}|y_i|} \times 100. \tag{3}$$

The best model is selected primarily according to the lowest wMAPE, while RMSE and R-squared are reported alongside it for transparency. If the total actual demand in the denominator is zero, wMAPE requires a defined fallback handling rule.

The model-selection procedure can therefore be summarized as: store-product series, train three models, generate test predictions, calculate metrics, select the model with the lowest wMAPE, and generate the final forecast. All three models' evaluation results are retained rather than discarded. This allows the user to inspect the model comparison and understand why the selected model was preferred.

VII. Inventory Recommendation

Forecasting demand alone does not directly solve the inventory decision problem. SalesSense therefore extends the forecasting output into a reorder recommendation.

The proposed system uses the predicted demand together with a configurable safety margin to estimate the recommended reorder quantity. A simplified representation is:

$$\text{Recommended Stock} = \text{Forecasted Demand} + \text{Safety Margin}. \tag{4}$$

The safety margin can provide additional inventory coverage against forecast uncertainty. This transforms the system from a forecasting-only application into a decision-support platform for inventory management.

VIII. System Architecture

SalesSense follows a three-tier architecture consisting of the presentation layer, application layer, and data and forecasting layer.

A. Presentation Layer

The presentation layer is implemented using React and provides a landing page, authentication, dashboard, dataset upload, model comparison, forecasting, inventory recommendations, and reporting. The frontend is responsible for rendering charts, KPIs, forecast results, and user interactions.

B. Application Layer

The backend uses FastAPI and acts as the interface between the frontend, database, and forecasting engine. Its responsibilities include request validation, authentication, data retrieval, forecast orchestration, communication with the forecasting engine, and returning forecasting results to the frontend.

C. Data and Forecasting Layer

MongoDB stores historical sales data, forecast results, model evaluation metrics, and user configuration. The Python forecasting engine contains the SARIMA, Prophet, and LSTM implementations and performs preprocessing, training, prediction, and evaluation.

IX. User Interface and Decision Support

SalesSense is designed around the principle that users should not need to be data scientists to obtain useful forecasting results. The primary workflow is:

1) Select a store.
2) Select a product or SKU.
3) Specify the forecast horizon.
4) View predicted demand.
5) View confidence information.
6) View the recommended reorder quantity.
7) Inspect model comparison when required.

The system provides a simplified primary result while retaining access to deeper model information. The best-performing model is displayed as the default result, whereas all three models can be compared side-by-side when the user wants additional information. The dashboard is designed to provide KPI cards such as predicted demand and stockout or overstock risk, historical sales trends, and exportable forecast reports.

X. Implementation Technology

The proposed technology stack is shown in Table I.

TABLE I
SalesSense Technology Stack

Component	Technology
Frontend	React
Backend	FastAPI
Database	MongoDB
Statistical Forecasting	statsmodels / SARIMA
Forecasting	Prophet
Deep Learning	TensorFlow / Keras
Dataset	FreshRetailNet-50K
Communication	REST API / HTTPS

The proposed implementation uses Python 3.11+, FastAPI, MongoDB, statsmodels, Prophet, and TensorFlow/Keras as the principal technologies.

XI. Results and Current Implementation Status

At the current stage of the project, the system architecture, dataset selection, requirements, user personas, model-selection methodology, and development roadmap have been defined. The FreshRetailNet-50K dataset has been finalized as the primary dataset, and the frontend has been developed as a prototype.

However, the current project documentation states that the forecasting models have not yet been trained or integrated, and that the backend has not yet been fully scaffolded. The existing frontend is currently operating with dummy or mock data. Therefore, quantitative forecasting results such as RMSE,

R-squared, or wMAPE values should not be reported as experimental results at this stage.

The planned evaluation will compare SARIMA, Prophet, and LSTM using the same store-product series and chronological test set. The final research results should report, for each evaluated series, the performance of all three models.

TABLE II
PLANNED FORECASTING MODEL EVALUATION

Series	Model	RMSE	R^2	wMAPE
Series 1	SARIMA	–	–	–
Series 1	Prophet	–	–	–
Series 1	LSTM	–	–	–

Once implementation is complete, this evaluation can demonstrate whether product-specific model selection improves forecasting performance compared with using a single model universally.

XII. DISCUSSION

The primary contribution of SalesSense is the use of product-specific model selection rather than applying a single forecasting algorithm across an entire retail catalog. Retail products can exhibit fundamentally different demand behavior. Stable products may contain strong recurring patterns, while promotional or irregular products may exhibit nonlinear fluctuations. The three-model architecture addresses this diversity by allowing different forecasting techniques to be preferred for different store-product combinations.

The inclusion of model comparison also improves transparency. Rather than simply presenting an unexplained prediction, the system retains the performance of all three candidate models and identifies the selected model based on measurable evaluation criteria. This approach is intended to reduce the perception of forecasting as a black-box process.

Another important aspect is the transition from prediction to action. By converting forecast demand into a reorder recommendation, SalesSense attempts to connect machine-learning output with an operational inventory decision.

Nevertheless, the current scope has limitations. The project evaluates only a curated subset of 5–10 store-product series because of the academic project timeline and team size. In addition, live point-of-sale integration, multi-tenant retailer support, real-time inventory integration, promotional what-if simulation, and stockout alerts remain outside the current implementation scope.

XIII. LIMITATIONS AND FUTURE WORK

The current SalesSense design contains multiple limitations. First, the evaluation is restricted to a relatively small subset of the FreshRetailNet-50K dataset. A more substantial assessment would be required to evaluate the generalizability of the model-selection approach to a significantly larger number of store-product combinations.

Second, the current design does not implement direct connectivity to a point-of-sale or inventory management system. Finally, the current design does not implement automated stockout alerts, promotional what-if simulation, and multi-store aggregation, which are intended for future implementation.

Future work can therefore include:

- expansion of model evaluation to a greater number of SKUs;
- addition of real-time point-of-sale and inventory data processing;
- automated stockout alert generation;
- multi-store aggregation;
- what-if promotional scenario simulation;
- improved handling of intermittent and sparse demand;
- long-term forecast accuracy tracking; and
- investigation of ensemble forecasting approaches.

The project roadmap also includes implementation of edge-case logic for new products and intermittent or sparse demand.

XIV. CONCLUSION

This paper introduced SalesSense, a proposed AI-powered retail demand forecasting and inventory recommendation platform that aims to provide accessible and explainable forecasting for retail users. It addresses the limitations of manual forecasting and the complexity of enterprise forecasting platforms by implementing three distinct forecasting approaches: SARIMA, Prophet, and LSTM.

Rather than assuming that a single approach is superior for all products, SalesSense uses chronological hold-out validation and compares the accuracy of each method using wMAPE, RMSE, and R-squared to select the best-performing model for a given store-product combination, with wMAPE serving as the primary selection criterion. The chosen model then forecasts future demand, which is subsequently transformed into a reorder recommendation with a configurable safety margin.

Backed by the React, FastAPI, Python, and MongoDB technology stack, the platform is designed to provide a modular foundation for future development. At the current proof-of-concept stage, the system's architecture, requirements, dataset, and forecasting methods have been defined, while actual model training and backend integration remain to be completed. Future experimental testing will therefore be necessary to assess forecasting accuracy and determine the practical benefits of the proposed model-selection approach.

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